

MME8215: Operations Research - Seminar

I. Linear Programming

Linear Programming (LP) is a classic technique in Operations Research and was invented independently several times during World War II. With the invention of the Simplex algorithm, by George Dantzig (1947), the models could also be optimized, i.e. the best solution values of the decision variables given an objective and a set of constraints.

Let us see some LP models: "Incredible Chairs" and "Incredible Chairs II".

1.1 Incredible Chairs

The company Incredible Chairs produces two different types of chairs, A and B, where one unit (pallet of chairs) of Chair A can be sold for a profit of 4 and one unit of Chair B can be sold for a profit of 6. The chairs are produced on three production lines:

1. Line 1 requires 2 hours to produce one unit of Chair A, and at most 14 hours can be used due to other production on the line.
2. Line 2 requires 3 hours to produce one unit of Chair B, and at most 15 hours can be used.
3. Line 3 requires 4 hours to produce one unit of Chair A and 3 hours to produce one unit of chair B, and at most 36 hours can be used.

Suppose that you are the production planner at the company, how will you plan the production of the chairs? The first important thing to identify is what you are in control of. In other words, what are the decisions you can make as a production planner? In this case, you can decide **how many** of each type of chair to produce. Since these quantities are unknown and must be decided, we define two decision variables, $x_A \geq 0$ and $x_B \geq 0$, which will store the quantity of each type of chair. Notice that the variables have a domain, i.e. an interval of feasible values. Here, they are each constrained to take a positive value. The job of the production planner is to find the best possible values for these two variables.

There are many possible solutions represented by the decision variables (x_A and x_B). As the production planner, you are, not interested in a random production plan, stated in the variables x_A and x_B , but you want the **best feasible** production plan. To compare the different solutions, an **objective function** $f(x_A, x_B)$ which quantifies the utility of the solution is used. For this example, the objective function measures the profit of the solution obtained by producing x_A and x_B chairs. The objective function can therefore be stated as:

$$\max f(x_A, x_B) = 4 \cdot x_A + 6 \cdot x_B.$$

Not all solutions are possible; they have to be **feasible**. This means that the values assigned to the variables should satisfy the restrictions that are imposed by the production lines. In particular, for each production line, the planned production of chairs cannot exceed the available resource limit. On line 3, the sum of the required resources cannot exceed 36. Given the unknown values of the decision variables, we can now formulate an equation - a **constraint** - that limits the values of x_A and x_B to values that are possible given the limitations on line 3:

$$4 \cdot x_A + 3 \cdot x_B \leq 36.$$

Similarly, there are two other constraints imposed by lines 1 and 2. They are given below:

$$2 \cdot x_A \leq 14$$

$$3 \cdot x_B \leq 15.$$

The complete LP model is therefore:

$$\begin{array}{ll} \text{Maximize} & 4 \cdot x_A + 6 \cdot x_B \\ \text{subject to} & \\ & 2 \cdot x_A \leq 14, \\ & 3 \cdot x_B \leq 15, \\ & 4 \cdot x_A + 3 \cdot x_B \leq 36, \\ & x_A \geq 0, \\ & x_B \geq 0. \end{array}$$

LP models are linear, i.e., the only allowed calculations in objective function and constraints are multiplications between constants and decision variables, and the addition and subtraction of variables.

Solve the problem.

1.2 Incredible Chairs 2

What if the company were not producing two different chairs on three production lines, but say ten different chairs on five different production lines? If the same approach were used, the size of the LP model would increase, and it would be quite tedious to input using the direct approach (1.1). In this case, we can formulate a more abstract model. Given a set of chairs C , indexed by c , and a set of production lines P , indexed by p , we can define a (one-dimensional) vector $Profit_c$, which states the profit for each chair $c \in C$, a (one-dimensional) production capacity vector $Capacity_p$, which states the production capacity of each production line $p \in P$, and a (two-dimensional) matrix $RecourseUsage_{p,c}$ that states how much of the available resource on production line p is needed for production of one unite of chair c .

This gives the following LP model:

$$\begin{array}{ll} \text{Maximize} & \sum_{c \in C} Profit_c \cdot x_c \\ \text{subject to} & \\ & \sum_{c \in C} RecourseUsage_{p,c} \cdot x_c \leq Capacity_p, \forall p \in P, \\ & x_c \geq 0, \forall c \in C. \end{array}$$

The above model is a very generic (chair) production model for profit maximization for any number of chairs and any number of production lines. In fact, any (simple) profit maximizing production problem with a number of products and production resources can be modeled in this way.

Write code that solves the generic problem.

1.3 Diet problem

We wish to cook a nutritionally balanced meal by choosing the quantity of each food f to eat from a set of foods F in our kitchen.

Each food f has a cost, c_f , as well as a macro-nutrient profile $a_{m,f}$ for each macro-nutrient $m \in M$.

Because we care about a nutritionally balanced meal, we set some minimum and maximum limits for each nutrient, which we denote l_m and u_m respectively.

Furthermore, because we are optimizers, we seek the minimum cost solution.

With a little effort, we can formulate our dinner problem as the following linear program:

$$\begin{aligned} & \text{Minimize } \sum_{f \in F} c_f x_f \\ & \text{subject to } l_m \leq \sum_{f \in F} a_{m,f} x_f \leq u_m, \quad \forall m \in M \\ & \quad \quad \quad x_f \geq 0, \quad \quad \quad \forall f \in F. \end{aligned}$$

Solve the diet problem.